

Copied centres in the two-stage interchange

Draft, October 9, 2026

This note applies PR #36's copied retained-centre schedule (icekylinx) to our two-stage bit interchange, where the centres are direct centre wires rather than retained totals. In that setting the schedule is simpler: the original centre never leaves its frame.

The central return. In each stage a centre c runs the dirty-scratch word $RGRG$: the scatter R reads c at the frame D_0 , and the gather G acts at D_1 . So the centre's frame path is $D_0 \rightarrow D_1 \rightarrow D_0 \rightarrow D_1$, with ranks h, h, h beyond its endpoint edge; this descent and re-ascent is the central-return loss, and it is what makes the rank budget $s = Wm - N + 2L$.

Lemma 1 (Copied centres). *In either stage, replace the second scatter's reads of c by reads of a temporary copy t : after the first gather, copy c into a fresh zero stream t at D_1 , move t to D_0 , let the second scatter read t , and erase t ; the original c stays at D_1 for the final gather. The scalar action, the endpoints of every role and the restoration of arbitrary scratch are unchanged. Each centre then has two children of width h per stage instead of three, and the rank budget becomes $s = Wm - N + L$.*

Proof. The scatter reads do not change c , so after the first gather the second scatter sees the same value whether it reads c or a copy of it; the final gather therefore receives the same value at the same frame D_1 . The copy is a fresh zero stream, so erasing it touches no arbitrary scratch. The move of t from D_1 to D_0 is the matrix of the old return edge, an involution with the same projector ($I \otimes P_t$ in stage one, $P_t \otimes I$ in stage two), so the flag basis compiles it to one run of width h , as before; the original's return and re-ascent, each of rank h , are gone. Summing over the $c_0 = h$ centres of the $2v$ invocations removes $L = 2vhc_0$ from the rank budget. $\square$

Tape cost. Making the copy is a pointwise gate on (c, t) at the common frame D_1 ; the reads are pointwise at D_0 ; erasure is linear in the stream volume V/W . The copy has the full role shape, so its child calls are ordinary children of the batched recursion, and $F_k \leq (s/W)F_{k-1} + O(1)$ holds with the new s . The copy is a temporary, not a role, so W is unchanged. The schedule needs $W+1$ role tapes, and while the copy's children run it parks W streams instead of $W-1$; both are fixed. No row of the assembly certificate changes: the bit network enters only through $\tau = 1 - a_b$, and the crude guard depends only on the complex network.

Saving. The deficit is now $Wm - s = N - L = v(v - 2h^2)$, positive for much smaller h , and the optimum moves from $h = 47$ to $h = 25$. With the flag basis, the gm side circuit ($R = 52788$ at $h = 25$), the side roles batched one level down and the data-entrance run, `scripts/certificate_round6.py` certifies $a_b = 34919/10^9$.

Retained totals. The direct centre wires can be replaced by retained point totals $P_c = \text{star}(c)$, built as side roles from existing side nodes (each a disjoint union of three of them, two additions). A total sits at U_c of dimension $h-1$; its copy pays $U_c \rightarrow D_0$ (rank $h-1$) and the original $U_c \rightarrow F$ (rank 1), so the loss per centre is $h-1$ and the centre roles leave W . At $h = 23$ this certifies $a_b = 36667/10^9$ (`independent/two-stage-bit/rtgm.py`, `cert_rt.py`); the scalar identity, frames, side-edge genericity and nesting are checked as for direct centres.

Checks. The full-stage word (all h centres, all targets, real excluded sums) is checked over $\mathbb{F}_2$ at $h = 5, \dots, 8$ and with signed Gaussian integers; a control that copies before the first gather fails. The frame sequences of old and copied centres are checked exactly at $h = 5, 6$. At $h = 25$ every flag-basis, side-lemma, data-entrance and frame condition is checked at one integer point, as at $h = 47$ in round five.